

Bachelor's Thesis Assignment



172897

Institut: Department of Computer Graphics and Multimedia (DCGM)
Student: **Kachan Rostyslav**
Programme: Information Technology
Title: **Large language models for solving mathematical problems**
Category: Speech and Natural Language Processing
Academic year: 2025/26

Assignment:

1. Familiarize yourself with the basics of neural networks for text generation.
2. Survey the state-of-the-art in large language models and their use in solving mathematical problems.
3. Prepare a suitable dataset for experimentation.
4. Design and implement a solution.
5. Perform experiments with your models.
6. Discuss the achieved results and possibilities of future development.
7. Create a poster presenting your work.

Literature:

- Touvron, Hugo, et al. "Llama: Open and efficient foundation language models." *arXiv preprint arXiv:2302.13971* (2023)

Requirements for the semestral defence:

Goals 1 to 3 and a prototype solution.

Detailed formal requirements can be found at <https://www.fit.vut.cz/study/theses/>

Supervisor: **Kostelník Martin, Ing.**
Head of Department: Černocký Jan, prof. Dr. Ing.
Beginning of work: 1.11.2025
Submission deadline: 13.5.2026
Approval date: 7.11.2025